Mathematics practice booklet: Grade 5 expectations

Grade 5 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 5 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 5.OA.A.1, 5.OA.A.2, 5.OA.B.3

A worked example

Work out 4 x (6 + 3) - 5. Then write the expression for: add 6 and 3, then multiply the sum by 4.

1. Brackets first: 6 + 3 = 9.
2. Then multiply: 4 × 9 = 36.
3. Then subtract: 36 - 5 = 31.
4. Expression for the words: 4 x (6 + 3).

31; 4 x (6 + 3)

Questions

1. What is 18 - (4 + 5)?

   A. 19

   B. 27

   C. 9

   D. 13

2. Evaluate: 3 x (12 - 7) + 8

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3. Evaluate: {20 - [3 x (2 + 1)]} / 2

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4. Which expression means three times the sum of 8 and 4?

   A. 3 × 8 + 4

   B. 3 x (8 + 4)

   C. 3 + 8 × 4

   D. (3 + 8) x 4

5. Without calculating, is 2 x (45 + 17) bigger or smaller than 45 + 17?

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6. Maryam buys 3 boxes of dates at AED 12 each and pays with a AED 50 note. Write one expression for her change and work it out.

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7. Hamdan puts AED 3 in his money box each day and Noura puts in AED 6 each day. They both start with nothing. Write the amount each has after 1, 2, 3, 4 and 5 days.

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8. Look at the money box amounts in the last question. How does Noura's amount compare with Hamdan's on each day?

   A. It is twice as big

   B. It is 3 more

   C. It is half as big

   D. It is the same

9. Write the ordered pairs (Hamdan's amount, Noura's amount) for the five days.

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10. A taxi charges AED 10 to start and AED 2 for each km. Which rule gives the cost in dirhams of a trip of k km?

    A. 10 x k + 2

    B. (10 + 2) x k

    C. 2 x k

    D. 10 + 2 x k

11. Insert brackets to make this true: 2 + 3 × 4 = 20

Section 2: Shapes, measuring and space

Standards practised: 5.MD.A.1, 5.MD.C.3, 5.MD.C.4, 5.MD.C.5, 5.G.A.1, 5.G.A.2, 5.G.B.3, 5.G.B.4

A worked example

A fish tank is 50 cm long, 30 cm wide and 40 cm high. Find its volume in cubic centimetres and in litres (1 litre = 1,000 cubic cm).

1. Volume = length x width x height.
2. 50 × 30 = 1,500, then 1,500 × 40 = 60,000 cubic cm.
3. 60,000 / 1,000 = 60 litres.

60,000 cubic cm, which is 60 litres

Questions

1. A gift box is 5 cm long, 4 cm wide and 3 cm high. What is its volume?

   A. 12 cubic cm

   B. 60 cubic cm

   C. 94 cubic cm

   D. 20 cubic cm

2. Layla stacks 1 cm cubes to build a bigger cube with edges of 4 cm. How many small cubes does she use?

   _____

3. A gift box has a volume of 120 cubic cm and its base is 10 cm by 4 cm. How tall is it?

   _____

4. Convert 2.5 m to centimetres.

   _____

5. Maha's family drinks 3,200 mL of water in a day. How many litres is that?

   A. 3.2 litres

   B. 32 litres

   C. 0.32 litres

   D. 320 litres

6. Point A is at (2, 3) on a grid. Which describes how to get there from the origin?

   A. Right 3, up 2

   B. Right 2, up 2

   C. Right 2, up 3

   D. Up 5

7. Plot the points (1, 1), (4, 1), (4, 3) and (1, 3) and join them in order. What shape do they make, and what is its area?

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8. A hotel pool is 10 m long and 5 m wide, and the water is 2 m deep. How many cubic metres of water fill it?

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9. Which statement is true?

   A. Every rectangle is a square

   B. Every rhombus is a square

   C. No square is a rectangle

   D. Every square is a rectangle

10. A four-sided shape has four equal sides, but its corners are not right angles. Name it. Is it a rectangle? Explain.

Section 3: Data, graphs and chance

Standards practised: 5.MD.B.2, 5.NF.A.1, 5.NF.B.3

A worked example

Water in 5 cups measures 1/2, 1/4, 1/4, 1/2 and 1/2 of a litre. If all the water is poured evenly into the 5 cups, how much does each hold?

1. Add: 1/2 + 1/4 + 1/4 + 1/2 + 1/2 = 2/4 + 1/4 + 1/4 + 2/4 + 2/4 = 8/4 = 2 litres.
2. Share equally between 5 cups: 2 / 5 = 2/5 of a litre.

2/5 of a litre

Questions

1. A line plot shows the karak in 6 cups: 4 cups hold 1/4 litre each and 2 cups hold 1/2 litre each. How many litres of karak is that in all?

   A. 3/4 litre

   B. 6 litres

   C. 1 litre

   D. 2 litres

2. Four cousins have 3, 5, 6 and 10 stickers. They put them together and share them equally. How many stickers does each get?

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3. A line plot shows 4 cups of water: 2 dots at 1/2 litre, 1 dot at 1/4 litre and 1 dot at 3/4 litre. How many litres of water are there in all?

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4. The water in the last question is poured evenly into the 4 cups. How much is in each cup?

   A. 1/2 litre

   B. 2 litres

   C. 1/8 litre

   D. 4 litres

5. In the worked example each cup holds 2/5 of a litre. Is that more or less than 1/2 litre?

   A. More, because 5 is greater than 2

   B. Less, because 2/5 = 4/10 and 1/2 = 5/10

   C. Equal

   D. It cannot be compared

6. Rangers at a mangrove reserve count eggs in nests on a line plot: 2 nests have 3 eggs each, 5 nests have 4 eggs each and 3 nests have 5 eggs each. How many nests are there?

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7. Using the same data, how many eggs are there in all?

   A. 12

   B. 60

   C. 41

   D. 40

8. In that data, how many eggs does a nest most often have?

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9. In the worked example, how much more water is in all the 1/2 litre cups than in all the 1/4 litre cups?

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10. Check the worked example: if each of the 5 cups holds 2/5 litre, how much water is that altogether? Does it match the total?

Section 4: Numbers and calculating

Standards practised: 5.NBT.A.1, 5.NBT.A.2, 5.NBT.A.3, 5.NBT.A.4, 5.NBT.B.5, 5.NBT.B.6, 5.NBT.B.7, 5.NF.A.1, 5.NF.A.2, 5.NF.B.3, 5.NF.B.4, 5.NF.B.5, 5.NF.B.6, 5.NF.B.7

A worked example

What is 2/3 + 1/4? What is 2/3 × 3/5?

1. Unlike denominators: find a common one, 12. 2/3 = 8/12 and 1/4 = 3/12.
2. 8/12 + 3/12 = 11/12.
3. Multiplying: top times top and bottom times bottom: 2 × 3 = 6 and 3 × 5 = 15, giving 6/15 = 2/5.

11/12 and 2/5

Questions

1. Write 4.075 in words. Then find 10 times 4.075 and one tenth of 4.075.

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2. Four samples of dates weigh 0.45 kg, 0.405 kg, 0.5 kg and 0.095 kg. Which is the heaviest?

   A. 0.45 kg

   B. 0.405 kg

   C. 0.5 kg

   D. 0.095 kg

3. The scale at the souq shows 7.846 kg of dates. Round the weight to the nearest hundredth of a kilogram.

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4. A trader packs 27 crates with 346 dates in each crate. How many dates is that?

   A. 2,422

   B. 9,342

   C. 6,920

   D. 8,342

5. A souq trader packs 1,248 dates into bags of 24. How many bags does he fill?

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6. Maryam buys 12.45 kg of dates and 3.8 kg of walnuts. How many kilograms is that in all?

   A. 12.83 kg

   B. 15.53 kg

   C. 15.25 kg

   D. 16.25 kg

7. Salem walks 1/2 km to the souq and then 1/3 km to the bakery. How far does he walk in all?

   A. 5/6 km

   B. 2/5 km

   C. 1/6 km

   D. 2/6 km

8. Layla has 3/4 litre of juice and drinks 1/6 litre. How much is left?

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9. A recipe uses 2/3 cup of flour. Maha makes 3/4 of the recipe. How much flour does she use?

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10. 4 cousins share 3 pizzas equally. What fraction of a pizza does each cousin get?

    A. 1/4

    B. 3/4

    C. 4/3

    D. 7/4

11. 3 friends share 1/4 of a pizza equally. What fraction of the whole pizza does each friend get?

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12. Without multiplying, is 5 × 2/3 greater or less than 5? Explain.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. C. 9. Brackets first: 4 + 5 = 9, then 18 - 9 = 9. If your child chose A (19), they ignored the brackets.
2. Brackets first: 12 - 7 = 5. Then 3 × 5 = 15, and 15 + 8 = 23.
3. Work from the inside out: 2 + 1 = 3, 3 × 3 = 9, 20 - 9 = 11, then 11 / 2 = 5.5.
4. B. 3 x (8 + 4) The sum 8 + 4 goes in brackets so it is added first. If your child chose A (3 × 8 + 4), they multiplied only the 8.
5. Bigger, because 2 times the sum means two copies of 45 + 17.
6. Expression: 50 - 3 × 12. Multiply first: 3 × 12 = 36, then 50 - 36 = AED 14.
7. Hamdan adds 3 each day: 3, 6, 9, 12, 15. Noura adds 6 each day: 6, 12, 18, 24, 30.
8. A. It is twice as big. 6 is 2 × 3 and 12 is 2 × 6, so every day is double. If your child chose B (3 more), they saw only day 1, and the gap grows.
9. Write each day as a pair, Hamdan first then Noura: (3, 6), (6, 12), (9, 18), (12, 24), (15, 30).
10. D. 10 + 2 x k. The 10 is paid once and the 2 is paid for each km. If your child chose B, (10 + 2) x k, they charged the start fee for every km.
11. Try brackets round 2 + 3: (2 + 3) x 4 = 5 × 4 = 20.

Section 2: Shapes, measuring and space

1. B. 60 cubic cm. Volume = 5 × 4 × 3 = 60. If your child chose A (12), they added the sides instead of multiplying.
2. 4 × 4 × 4 = 64: each layer has 16 cubes, and there are 4 layers.
3. Base area = 10 × 4 = 40. Height = volume / base area = 120 / 40 = 3 cm.
4. 1 m is 100 cm, so 2.5 × 100 = 250 cm.
5. A. 3.2 litres. 1 litre is 1,000 mL, so 3,200 / 1,000 = 3.2. If your child chose B (32 litres), they moved the decimal point only 2 places.
6. C. Right 2, up 3. The first number is across and the second is up. If your child chose A (right 3, up 2), they swapped the two numbers.
7. It is a rectangle, 3 units wide (4 - 1) and 2 units tall (3 - 1), so area = 3 × 2 = 6 square units.
8. Volume = length x width x depth = 10 × 5 × 2 = 100 cubic metres.
9. D. Every square is a rectangle. A square has four right angles, which is all a rectangle needs. If your child chose A (every rectangle is a square), they forgot that a rectangle can have unequal sides.
10. Four equal sides make a rhombus. A rectangle needs four right angles, and these corners are not right angles, so it is not a rectangle.

Section 3: Data, graphs and chance

1. D. 2 litres. 4 × 1/4 = 1 and 2 × 1/2 = 1, so 1 + 1 = 2. If your child chose B (6 litres), they counted the cups instead of adding the amounts.
2. 3 + 5 + 6 + 10 = 24 stickers altogether, then 24 / 4 = 6 each.
3. 2 × 1/2 = 1, then 1 + 1/4 + 3/4 = 2 litres.
4. A. 1/2 litre. 2 litres shared among 4 cups is 2 / 4 = 1/2. If your child chose B (2 litres), they gave the total, not one cup's share.

5. B. Less, because 2/5 = 4/10 and 1/2 = 5/10. Less. Make the bottoms the same: 2/5 = 4/10 and 1/2 = 5/10, and 4/10 is less. If your child chose A (more), they thought 5 is greater than 2, so the fraction is bigger.
6. Add the nests: 2 + 5 + 3 = 10.
7. C. 41. 2 × 3 = 6, 5 × 4 = 20 and 3 × 5 = 15, and 6 + 20 + 15 = 41. If your child chose A (12), they added 3 + 4 + 5 and forgot how many nests have each number.
8. 5 nests have 4 eggs, which is more nests than the 2 with 3 eggs or the 3 with 5 eggs, so 4 is most common.
9. 3 × 1/2 = 6/4 and 2 × 1/4 = 2/4. Then 6/4 - 2/4 = 4/4 = 1 litre.
10. 5 × 2/5 = 10/5 = 2 litres, which matches the total of 2 litres.

Section 4: Numbers and calculating

1. Say it as four and seventy-five thousandths. Times 10 moves each digit one place left (40.75), and one tenth moves each digit one place right (0.4075).
2. C. 0.5 kg. Compare the tenths first: 0.5 has 5 tenths and 0.45 has 4. If your child chose D (0.095), they thought more digits means a bigger number.
3. The thousandths digit is 6, which is 5 or more, so round 7.84 up to 7.85 kg.
4. B. 9,342. 346 × 20 = 6,920 and 346 × 7 = 2,422, so 6,920 + 2,422 = 9,342. If your child chose C (6,920), they multiplied by 20 only and forgot the 7.
5. 24 × 50 = 1,200, and the 48 left is 2 more bags of 24, so 52 bags.
6. D. 16.25 kg. Line up the decimal points: 12.45 + 3.80 = 16.25. If your child chose A (12.83), they lined up the last digits instead of the decimal points.
7. A. 5/6 km. Use sixths: 1/2 = 3/6 and 1/3 = 2/6, so 3/6 + 2/6 = 5/6. If your child chose B (2/5), they added the tops and added the bottoms.
8. 3/4 = 9/12 and 1/6 = 2/12, so 9/12 - 2/12 = 7/12 litre.
9. Multiply the tops and the bottoms: 2/3 × 3/4 = 6/12, which simplifies to 1/2.
10. B. 3/4. Sharing 3 pizzas among 4 cousins is 3 / 4 = 3/4 of a pizza each. If your child chose C (4/3), they divided 4 by 3 instead.
11. 1/4 divided by 3 is 1/4 × 1/3 = 1/12, so each friend gets 1/12 of the pizza.
12. Less than 5, because 2/3 is less than 1, so 5 groups of 2/3 are smaller than 5 groups of 1.